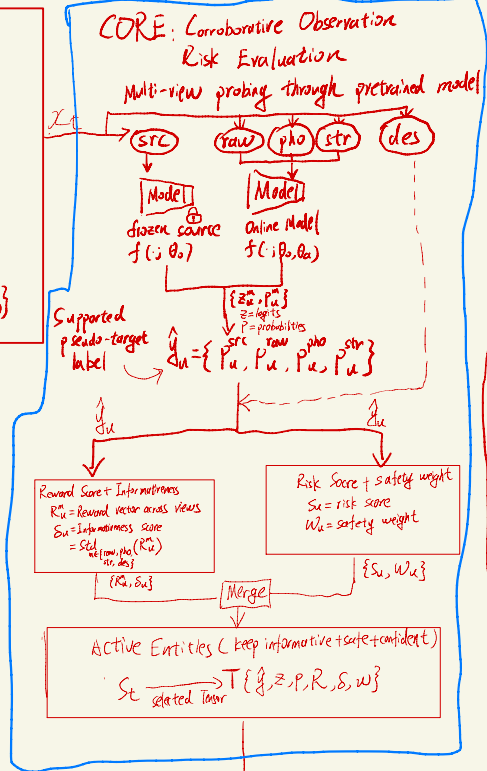


Input
 unlibed
 target batch
 $\mathcal{X}_t \in \mathcal{R}^{B \times C_u \times H \times W}$
 $\{\text{image}/\text{pixel}/\text{prompt-view}\}$



CARE: Conservative Anchored Ratio Estimation

clean policy:
 BN/GN/VLM $\rightarrow$ source-relative
 ViT-LN cls/seg $\rightarrow$ class-centered.

Source-relative anchor
 r_u = source-relative ratio
 A_u = raw-view reliability
 $C_u^{\text{src}} = T\{g, p^{\text{raw}}, p^{\text{src}}, R, S\} = \phi(r_u, A_u)$

class-centered anchor
 $C_k^{\text{src}} \in \mathcal{R}^s$ is class center
 $C_u^{\text{src}} = T\{g, z^{\text{raw}}, p^{\text{raw}}\} + C_k^{\text{src}}$
 $= \hat{p}_u^T z_u^{\text{raw}}$

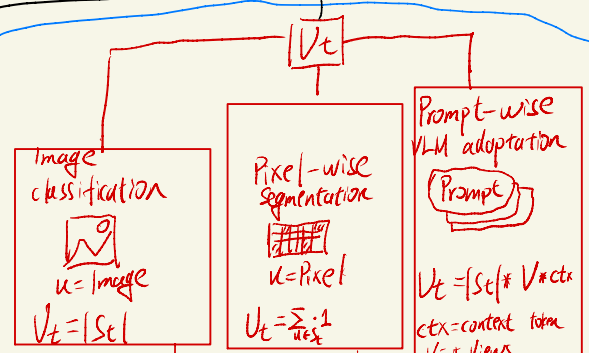
$CARE_u = \begin{cases} \phi(r_u, A_u), & \text{if BN/GN/VLM} \\ \hat{p}_u^T z_u^{\text{raw}}, & \text{if ViT-LN cls/seg} \end{cases}$

Inference

Online Model
 $f(\cdot; \theta_o, \theta_u)$
 Update θ_u

Total loss

$$\mathcal{L}_t^{\text{atlas}} = -\frac{1}{U_t} \sum_{u \in U_t} \mathbb{1}[u \in \mathcal{S}_t] \cdot W_u \cdot C_u^{\text{src}}$$



SANE: Scale-Aware Normalization of Entities